

Anderson Junior College
Preliminary Examination 2009
H2 Mathematics Paper 2

Section A: Pure Mathematics [40 marks]

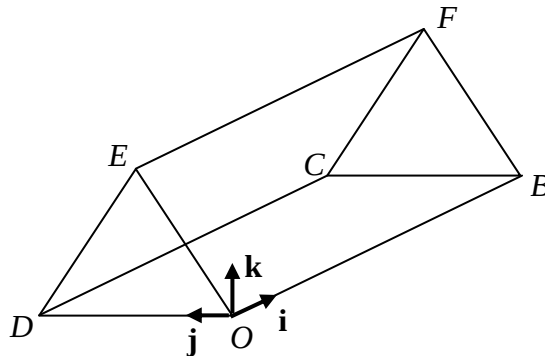
1. A curve with equation $y = f(x)$ passes through the point $(0,1)$ and satisfies the relation

$$(1+x^2)\frac{dy}{dx} + xy = \sqrt{1+x^2}.$$

By further differentiation of this result, find the Maclaurin's series for y in ascending powers of x , up to and including the term in x^2 . [3]

Deduce the approximate value of $\int_0^{0.01} \frac{dy}{dx} dx$, giving your answer correct to 3 decimal places. [2]

2. The diagram shows a roof, with horizontal rectangular base $OBCD$, where $OB = 10\text{m}$ and $BC = 6\text{m}$. The triangular planes ODE and BCF are vertical and the ridge EF is horizontal to the base. The planes $OBFE$ and $DCFE$ are each inclined at an angle θ to the horizontal, where $\tan \theta = \frac{4}{3}$. The point O is taken as origin and vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, each of length 1m , are taken along OB , OD and vertically upwards from O respectively.



- (i) Show that $\overrightarrow{OF} = 10\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$. [2]
- (ii) Find the acute angle between the line OF and the plane BDE , giving your answer to the nearest 0.1° . [4]
3. (a) By using the substitution $z = x + y$, show that the differential equation $(x+y)\frac{dy}{dx} = x^2 + xy + x + 1$ can be reduced to the equation $z\frac{dz}{dx} = (x+1)(z+1)$. Hence find the general solution of the differential equation

$$(x+y)\frac{dy}{dx} = x^2 + xy + x + 1. \quad [4]$$

3. (b) John jogs 8 km every morning. As a mathematical model, he assumes that his speed during the jog is proportional to the distance he has yet to complete.
- (i) He starts off at an initial speed of 10 km per hour, and after t hours, he has jogged x km. Show that $\frac{dx}{dt} = \frac{5}{4}(8-x)$. [1]
- (ii) Find an expression for x in terms of t and hence find the time taken for John to complete 6 km. [3]
- (iii) Comment on the suitability of the model. [1]

4. A sequence of positive real numbers $x_1, x_2, x_3, \dots$ satisfies the recurrence relation

$$x_{n+1} = \frac{k + 2x_n}{2 + x_n}, \quad n \in \mathbb{Z}^+, \text{ where } k \text{ is a positive constant.}$$

- (i) Given that the sequence converges to α , find α in terms of k . [2]
- (ii) By considering $x_{n+1} - x_n$, show that if $x_n < \sqrt{k}$, then $x_{n+1} > x_n$. [2]
- (iii) Prove that $x_{n+1} - \sqrt{k} = \frac{(x_n - \sqrt{k})(2 - \sqrt{k})}{2 + x_n}$.

Hence find the range of values of k for which $x_{n+1} < \sqrt{k}$ whenever $x_n < \sqrt{k}$ for all $n \in \mathbb{Z}^+$. [3]

- (iv) If $k = 1$ and $x_1 = 0.9$, find x_2 and x_3 correct to 3 decimal places. [1]
- (v) Describe the behaviour of the sequence obtained in part (iv). [2]

5. (a) Solve the simultaneous equations

$$iz + 2w = 1 \text{ and } 4z + (3-i)w^* = -6,$$

giving z and w in the form $a + bi$ where a and b are real. [5]

- (b) (i) Find the roots of the equation $z^5 = 1$, giving your answers exactly in the form $re^{i\theta}$. [1]

(ii) Hence show that the roots of the equation $w^5 - (w-i)^5 = 0$, $w \neq i$ are

$$\frac{1}{2} \left(\cot \frac{k\pi}{5} + i \right), \text{ where } k = 1, 2, 3, 4, 5. \quad [4]$$

Section B: Statistics [60 marks]

6. In a ceremony, ten flags, consisting of 3 red, 5 blue and 2 yellow, are arranged in a straight line for decoration. The flags are identical apart from the colour. Find the number of ways in which the flags can be arranged if
- (i) there are no restrictions, [1]
 - (ii) at least 3 blue flags are together. [4]

At the end of the ceremony, the flags are numbered from 1 to 10. Four out of the ten flags are to be chosen randomly and presented together to the Guest of Honour. How many selections are possible if at most one yellow flag is included? [2]

7. A full set of 52 poker cards consists of 4 suits, Heart, Club, Diamond and Spade. A set of 40 cards is formed by removing all Jacks, Queens and Kings from a full set of 52 poker cards.

- (a) Using the set of 40 cards, two cards are drawn at random without replacement.
 - (i) Show that the probability of drawing at least 1 diamond is $\frac{23}{52}$. [2]
 - (ii) If at least one diamond is drawn, find the probability of getting exactly one '7'. [3]
- (b) Suppose the set of 40 cards is reshuffled and a card is drawn at random. The number on the card is noted. If the card is not a '7', it is replaced and another card is drawn at random. The procedure is repeated until a '7' is drawn. Find the greatest value of n such that the probability of obtaining the first '7' on or after the n^{th} draw is at least 0.4. [3]

8. A company has 400 employees in 3 departments, Administrative, Finance and Marketing. The employees have 2 different skill levels, Level 1 and Level 2. The number of employees in each department is classified by skill level and gender according to the table below:

Department	Administrative		Finance		Marketing	
Skill Level	Level 1	Level 2	Level 1	Level 2	Level 1	Level 2
Male	25	20	13	51	7	55
Female	45	22	18	72	12	60

A survey on the employees' attitude to a new pay scheme is to be carried out on a sample of 50 employees.

- (i) Explain why a systematic random sampling method may not be appropriate. [1]
- (ii) Explain (but do not calculate) how you would construct a stratified sample representing all groups in the company. [1]

The company director decides that the stratified sample shall reflect only gender and skill level differences. Show how such a sample of 50 employees can be obtained. [2]

9. The manager of a bank receives, on average, two complaints of poor service from its customers per week.
- (i) Given that the manager receives 3 complaints in a month, find the probability that he receives all of them in the first three weeks of the month. (Assume that there are 4 weeks in a month). [2]
 - (ii) Find the probability that, in a random sample of six weeks, the manager receives no complaint in one week, one complaint each in three weeks and more than one complaint in each of the remaining two weeks. [3]
 - (iii) In a randomly chosen period of n weeks (where $n > 5$), the probability that the manager receives more than $3n$ complaints is less than 0.01. Using a suitable approximation, find the least value of n . [4]
10. Customers at a supermarket pay for their purchases either by cash or credit card. The probability of a randomly chosen customer paying by credit card is 0.3.
- (i) Find the probability that, on a randomly chosen day, there are more than 6 customers among the first 20 customers who pay by credit card and the fifth customer is the second customer who does so. [3]
 - (ii) Fifty samples of 15 customers each were taken. If there is a 95% chance that the average number of customers who pay by credit card per sample lies in the interval $(4, a)$, find the value of a , giving your answer to 3 significant figures. [3]
11. A resident of a house near a road claims that the average speed of vehicles along the road is greater than the speed limit of 60 km per hour. The town council recorded the speed, x km per hour, of 20 randomly selected vehicles and summarised the data collected as follows:

$$\sum (x - 60) = 62 \quad \sum (x - 60)^2 = 1096.$$

It is assumed that the speed of the vehicles is normally distributed.

- (i) State appropriate null and alternative hypotheses for a significance test. [1]
- (ii) Test at 5% level of significance whether there is evidence to support the resident's claim. [4]
- (iii) Explain what is meant by "5% level of significance" in the context of this question. [1]

It is now known that the speed of the vehicles is normally distributed with mean μ km per hour and standard deviation 24 km per hour. To investigate the resident's claim, the police recorded the speed of 25 randomly selected vehicles.

- (iv) Find the range of values for the sample mean $\bar{x}$ such that there is evidence to support the resident's claim at the 5% level of significance. [2]
- (v) State, with a reason, your conclusion for the test if the mean speed calculated from the sample was 65 km per hour. [1]

12. One end A of an elastic string was attached to a horizontal bar and a mass, m grams, was attached to the other end B . The mass was suspended freely and allowed to settle vertically below A . The length AB , l mm, was recorded, for various masses as follows:

m	100	200	300	400	500	600
l	228	236	256	a	285	301

The equation of the regression line of l on m is $l = 0.15257m + 210.6$.

- (i) Show that $a = 278$, corrected to the nearest integer, and find the product moment correlation coefficient of l and m . [2]
- (ii) Give, in context, interpretations for the gradient and the vertical intercept of the regression line of l on m . [2]
- (iii) State a possible physical limitation in using your regression line of l on m to estimate the length of the string when a mass of 1200 grams is attached at B . [1]

Using the above table, a statistician derived a new set of data with 2 variables M and L where $M = \frac{m}{100}$ and $L = l - b$, b is a constant. Find

- (iv) the regression coefficient of L on M . [2]
- (v) the product moment correlation coefficient of L and M . [1]

13. A telco company found that the duration of calls made by its customers to City A are normally distributed with mean 8 minutes and standard deviation 2.5 minutes. The duration of calls made by its customers to City B are normally distributed with mean 10 minutes and standard deviation 3.2 minutes. The duration of any call is independent of the duration of any other calls.

- (i) Find the probability that the total duration of two randomly selected calls to City A differs from thrice the duration of a third call to City A by more than three minutes. [3]

The company charges a rate of 22 cents per minute for every call to City A and a rate of 30 cents per minute for every call to City B . For each call, customers are also charged a handling fee of 10 cents. Ann makes two calls to City A and one call to City B .

- (ii) Find the total amount of money she is expected to pay for the calls. [1]
- (iii) Show that the probability that she spends more than \$8.50 on the calls is 0.0870, corrected to 3 significant figures. [2]

Over a period of 50 weeks, Ann makes two calls to City A and one call to City B every week. By using a suitable approximation, find the probability that she spends more than \$8.50 in more than three but less than eight of the weeks. [3]

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